

Stochastic Differential Equations

Itô vs Stratonovich – Simulation & Analysis

Numerical simulation and analytical verification of two geometric Brownian motion SDEs under different stochastic calculus conventions, using the Euler–Maruyama scheme.

SDEs Under Study

	SDE	Convention	Itô-equivalent form
B1	$dX = X \, dW$	Itô	$dX = X \, dW$
B2	$dX = X \circ dW$	Stratonovich	$dX = \frac{1}{2}X \, dt + X \, dW$

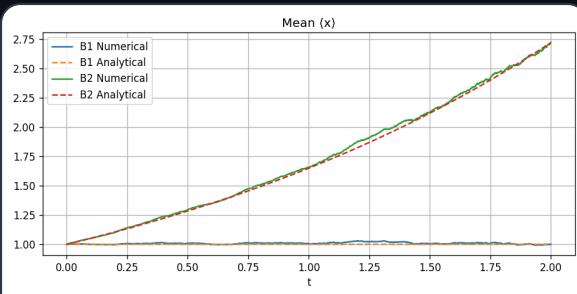
Both start at $X(0) = 1$. The extra $\frac{1}{2}X \, dt$ in B2 is the Itô–Stratonovich correction term.

b.py – Moment Verification

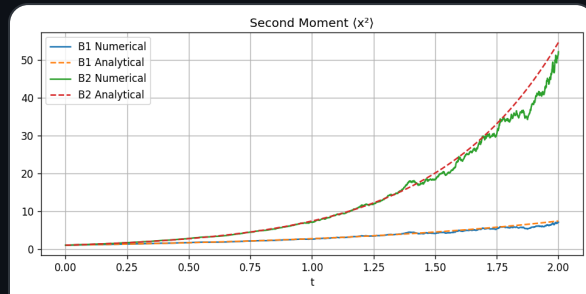
Simulates 5,000 trajectories up to $T = 2.0$ and checks numerical moments against closed-form analytical solutions.

Analytical Moments

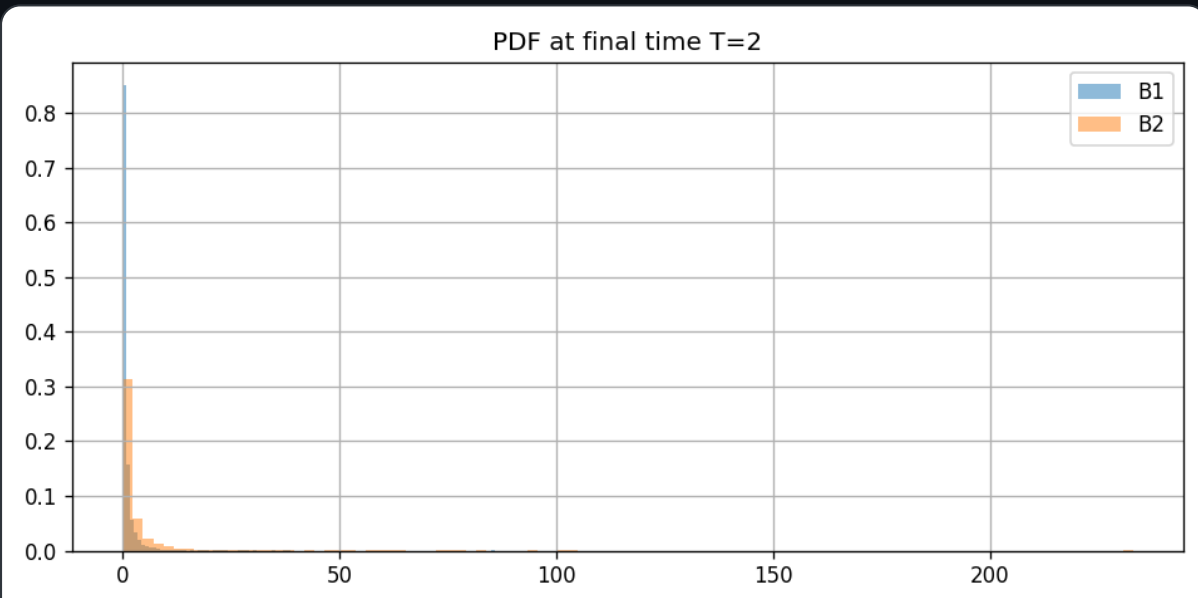
Moment	B1 (Itô)	B2 (Stratonovich)
$\langle x \rangle$	1	$e^{(t/2)}$

$\langle x^2 \rangle$ e^t $e^{(2t)}$ 

Mean $\langle x \rangle$ vs time – B1 stays flat at 1 (Itô martingale), B2 grows as $e^{(t/2)}$



Second moment $\langle x^2 \rangle$ vs time – B1: e^t , B2: $e^{(2t)}$



Empirical PDF at final time $T = 2$ – B2 is heavier-tailed and more spread out

B1's mean is constant (Itô martingale property). B2 grows exponentially in mean and variance – a direct consequence of the Stratonovich correction adding a positive drift.

a2.py – PDF Evolution

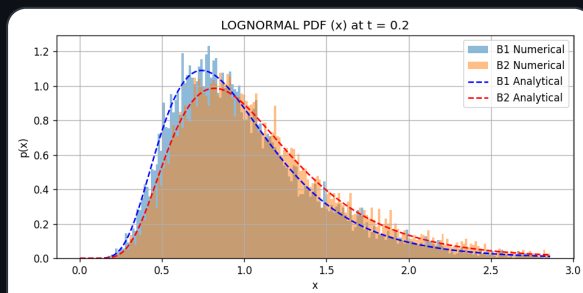
Simulates **10,000 trajectories** and tracks the full probability distribution at four snapshots: $t = 0.2, 0.5, 1.0, 2.0$.

Analytical Distribution

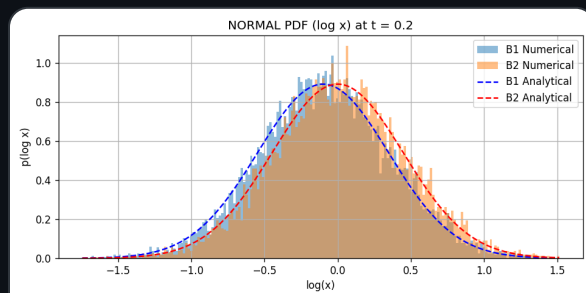
Both B1 and B2 are lognormally distributed. Writing $X = e^Y$ where Y is Gaussian:

	$\log(X)$ mean	$\log(X)$ variance
B1	$-t/2$	t
B2	0	t

$t = 0.2$ – Early time

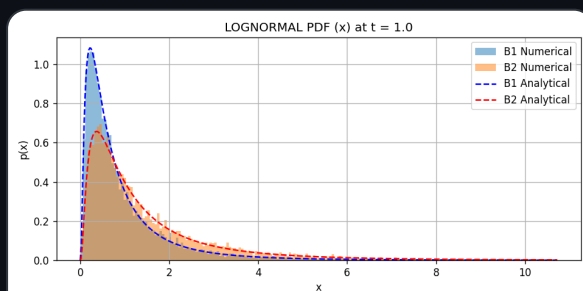


Lognormal PDF (x-space) at $t = 0.2$

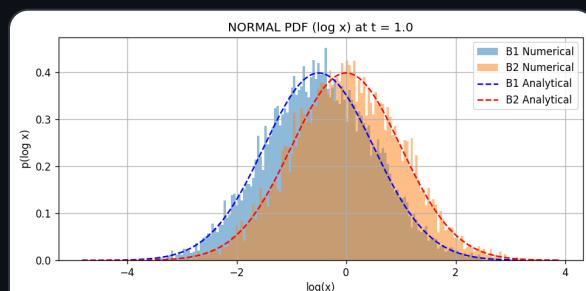


Normal PDF (log x-space) at $t = 0.2$

$t = 1.0$ – Midpoint

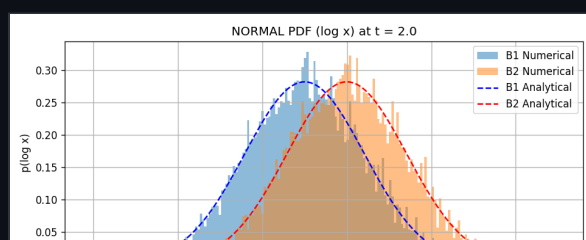
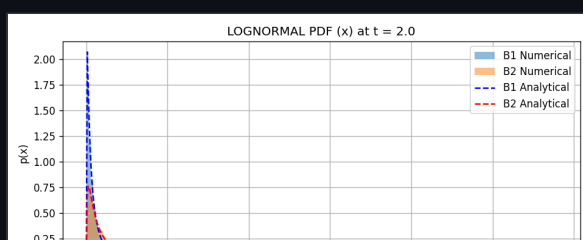


Lognormal PDF (x-space) at $t = 1.0$



Normal PDF (log x-space) at $t = 1.0$
– Gaussian structure clearly visible

$t = 2.0$ – Final time



Lognormal PDF (x-space) at $t = 2.0$
– heavy right tail emerges

Normal PDF (log x-space) at $t = 2.0$
– B1 drifted left, B2 centred at 0

The log-space plots confirm both processes remain Gaussian in $\log(x)$ at all times. B1 drifts to $-t/2$ while B2 stays centred – same spread, different location.



Running the Scripts

```
pip install numpy matplotlib
```

```
python b.py      # Moment verification (M=5000, T=2.0)
```

```
python a2.py     # PDF evolution      (M=10000, T=2.0)
```



Key Concepts

Concept	Description
Itô integral	Integrand evaluated at <i>left</i> endpoint – yields martingale property for B1
Stratonovich integral	Integrand evaluated at <i>midpoint</i> – consistent with classical chain rule
Itô–Stratonovich correction	The extra $\frac{1}{2}X \, dt$ drift in B2 that converts Stratonovich to Itô form
Euler–Maruyama	Discretisation: $X_{i+1} = X_i + \text{drift} \cdot dt + X_i \cdot dW$, $dW \sim N(0, dt)$